

STEM Exploration for High Schoolers

Chapter 10

Worksheet

Instructions:

Please complete all activities in this worksheet. Write your answers in the spaces provided or on separate sheets if needed.

Worksheet: Biomedical Science and Engineering Adventure

Hey Health Heroes! Get ready to dive into the world of Biomedical Science and Engineering with this interactive worksheet. This isn't just busywork—it's a mission to explore how science and tech save lives! From cracking body system codes to designing med-tech gadgets, you'll test your knowledge, spark creativity, and solve real-world health puzzles. Grab a pen, team up with a friend if you want, and let's heal the world one STEM idea at a time. Post your coolest answers or sketches on the class board with #BioTechBoss for squad hype!

Part 1: Crack the Code - Body Systems Match-Up (Knowledge Check)

Mission: Match each body system to its main function and a real-world health glitch it might face. Draw a line or write the letter of the function and glitch next to the system. Bonus: Pick one system and doodle a quick "superhero" version of it (like "Heart Man" pumping power) in the margin!

Body System	Function Options	Health Glitch Options
● 1. Circulatory System	● A. Controls thoughts and actions	● X. Asthma (tight airways)
● 2. Respiratory System	● B. Pumps blood and oxygen	● Y. Diabetes (sugar imbalance)
● 3. Nervous System	● C. Takes in oxygen via breathing	● Z. Anxiety (overactive worry)

Body System

Function Options

Health Glitch Options

- 1. Circulatory System
- A. Controls thoughts and actions
- X. Asthma (tight airways)
- 2. Respiratory System
- B. Pumps blood and oxygen
- Y. Diabetes (sugar imbalance)
- 3. Nervous System

- C. Takes in oxygen via breathing
- Z. Anxiety (overactive worry)

Your Answers:

- Circulatory System: Function ____ | Glitch ____
- Respiratory System: Function ____ | Glitch ____
- Nervous System: Function ____ | Glitch ____

Reflection: Which system's health glitch sounds toughest to deal with, and why? Write 1-2 sentences.

Part 2: Med-Tech Mystery - Guess the Tool (Critical Thinking)

Mission: Read the clues about medical technologies and guess which tool is being described. Circle the correct answer from the options. Bonus: Add a quick "upgrade" idea for one tool (like "add a game to MRI to calm kids!") next to your answer.

- **Clue:** I use radiation to snap pictures of your bones—perfect for spotting a break after a fall. What am I?
 - A) MRI
 - B) X-Ray
 - C) Prosthetic **Your Guess:** _____ | Upgrade Idea: _____
- **Clue:** I'm a wearable gadget that tracks your heart rate and steps, sometimes even calling for help if you fall. What am I?
 - A) Ultrasound
 - B) Smartwatch
 - C) Stethoscope **Your Guess:** _____ | Upgrade Idea: _____
- **Clue:** I'm a replacement limb with sensors, helping someone run again after an injury. What am I?
 - A) Prosthetic
 - B) Insulin Pump
 - C) MRI **Your Guess:** _____ | Upgrade Idea: _____

Reflection: Which tool do you think helps people the most day-to-day, and why? Write 1-2 sentences.

Part 3: Bio-Engineer Brainstorm - Design a Health Gadget (Creativity)

Mission: You're a biomedical engineer tasked with solving a teen health issue (like stress, sports injuries, or acne). Pick one issue, name your gadget, describe what it does, and sketch it (or describe the sketch if you can't draw). Bonus: Add a hashtag for your invention to go viral!

- **Health Issue:** _____ (e.g., stress from exams)
- **Gadget Name:** _____ (e.g., ChillVibe Band)
- **What It Does:** Describe in 2-3 sentences how it helps. (e.g., "This wristband senses fast heart rate from stress, plays calming music through earbuds, and sends a 'breathe' reminder to your phone.")
- **Quick Sketch or Description:** Draw in the box below or write what it looks like (e.g., "small black band with a tiny screen and earbud jack").
- **Viral Hashtag:** # _____ (e.g., #ChillVibeFix)

Reflection: What's one challenge (like cost or size) your gadget might face in real life? Write 1-2 sentences.

Part 4: Ethics Dilemma Debate - What Would You Do? (Ethical Thinking)

Mission: Biomedical Science often faces tough choices. Read the scenario below, pick a side, and write a short argument (3-4 sentences) for your decision. Bonus: Share your stance with a partner and see if they agree—jot

down one thing you learned from their view!

Scenario: A new gene-editing tool can cure a deadly disease, but it's super expensive—only 1% of patients can afford it. Should it be released now to help a few, or delayed until it's cheaper for everyone?

- **Your Choice:** ☐ Release Now ☐ Delay Until Cheaper
- **Your Argument:** Why did you pick this side? Explain your reasoning.
- **Partner Chat Bonus:** What's one point your partner made (if you discussed)?

Reflection: How did thinking about fairness in medicine make you feel? Write 1-2 sentences.

Part 5: Career Quest - Map Your Future (Exploration)

Mission: Explore a career in Biomedical Science or Engineering that excites you. Pick one from the list (or make up your own), answer the questions, and imagine your “dream day” on the job. Bonus: Create a cool job title sticker (like “Gene Genius”) to slap on your notebook!

- **Career Choice:** _____ (e.g., Biomedical Engineer, Genetic Counselor, Health App Developer)
- **What They Do:** Write 1-2 sentences about their main role. (e.g., “Designs medical devices like prosthetics to help people move again.”)
- **Skill I'd Need:** Name one STEM skill to start learning now. (e.g., “Coding for app design.”)
- **Dream Day on the Job:** Describe one epic moment in a day. (e.g., “Testing a new robotic arm with a patient and seeing them high-five for the first time!”)
- **Cool Job Title Sticker:** Write your custom title. (e.g., “Robo-Limb Rockstar”) #_____

Reflection: What's one small step (like a club or online video) you could take this month to learn more about this career? Write 1-2 sentences.

Bonus Challenge: Bio-Tech Trivia Throwdown

Mission: Test your squad's knowledge! Write one trivia question about anything from Chapter 10 (body systems, med-tech, ethics, etc.) with the answer. Challenge a friend or the class to answer it. Post it with #BioTechTrivia for extra clout!

- **My Trivia Question:**
- **Answer:**